The Palaeomagnetism of the Stormberg Lavas, II. The Behaviour of the Magnetic Field During a Reversal

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Summary

The main theme of this paper is to investigate the intensity of the ancient geomagnetic field during the transition from a reversed to a normal direction as the Stormberg lavas were being extruded in Triassic–Jurassic times.

The nature of the natural remanent magnetization of these rocks is investigated. It is thought that the effects of secondary magnetization, notably a high isothermal remanent magnetization believed to be due to lightning discharges and a remanent component resembling viscous magnetization, have been eliminated.

The direction and intensity of magnetization of the normal and reversed zones are described and compared. It seems probable that the field strength decreased by a factor of four to five during the transition.

1. Introduction

This paper is the second on the palaeomagnetism of the Stormberg lavas. The first paper described evidence strongly supporting the hypothesis that the reversal of direction of magnetization between the lowermost lavas and those higher in the succession was due to a reversal of the geomagnetic field during the time of extrusion of the lavas.

The initial measurements of the intensity of magnetization (J), the declination (D) and the inclination (I) together with α 99 per cent (Fisher 1953) are shown in Figure 1. It will be noted that the intensity curves in Figure 1 suggest that J decreases in the transition zone. In view of the importance attached to the elucidation of the behaviour of the geomagnetic field during a reversal, it was considered worthwhile to make more detailed studies of the intensities. For this reason detailed investigations of the nature of the natural remanent magnetization were made in order to establish whether (i) the natural remanent magnetization (NRM) was a thermoremanent magnetization (TRM); or (ii) any magnetization other than the original TRM could be detected or removed.

2. The investigation of the nature of the N.R.M.

(*a*) *The D and I curves.*—The overall stability of the magnetization of the lavas in the reversed and transition zones is suggested by the fact that:

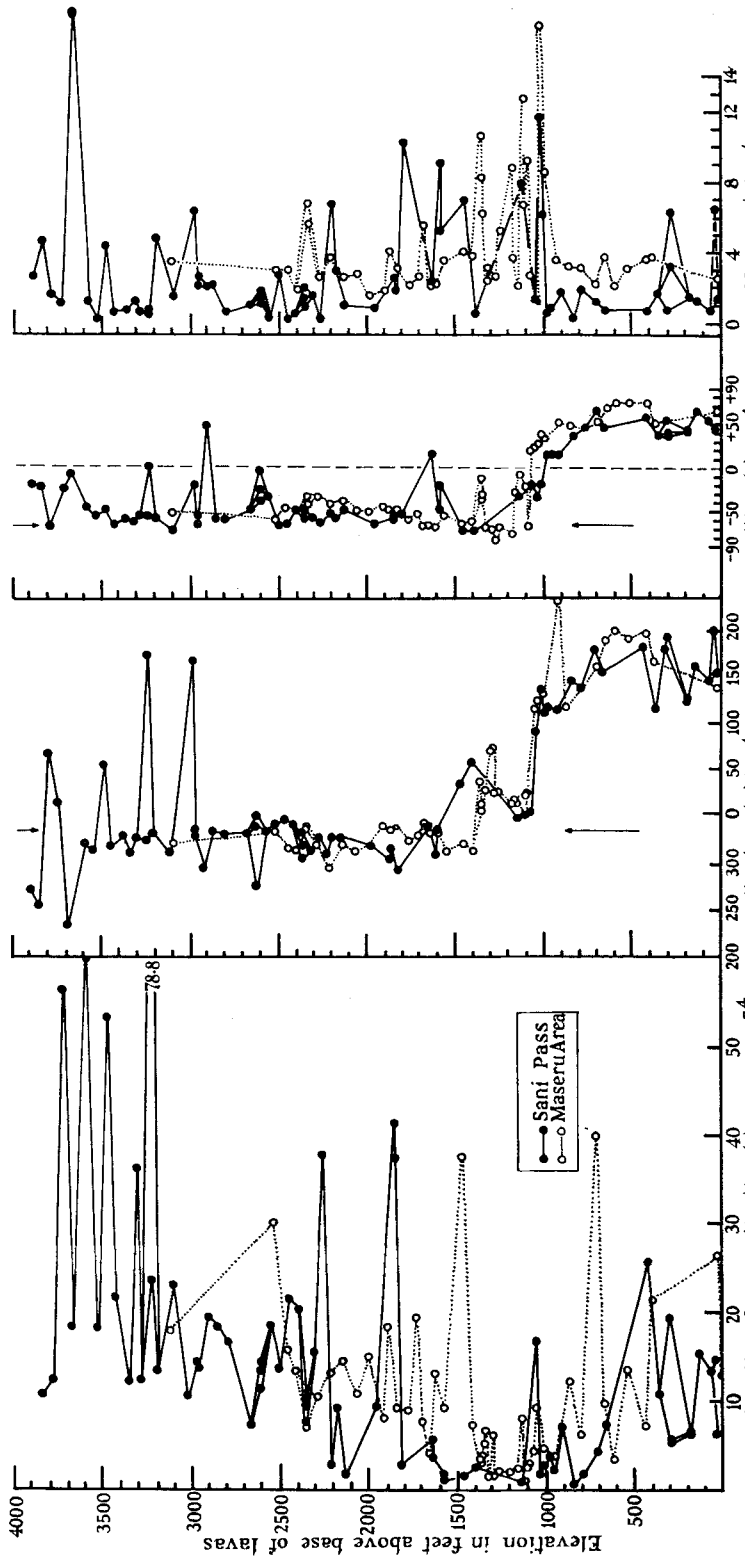


FIG. 1.—The intensity (J), declination (D), inclination (I) and the radius of the cone of 99 per cent confidence (α 99 per cent) of the mean direction of magnetization of the cores from Sani Pass and the Maseru Area as originally measured, plotted against elevation in feet above the base of the lavas. Some flows which gave apparently anomalous results were resampled. In such cases more than one point is plotted at a particular elevation. The arrows indicate the declination and inclination of the present field.

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(i) The D and I values for the transition and reversed zones (Figure 1) differ appreciably from that of the present field direction; and

(ii) the elevation in the lava succession at which the transition takes place corresponds closely at the two sampling areas which are 80 miles apart.

(*b*) *The ratio NRM/TRM.*—In the J curves of Figure 1 irregular high peaks are noted especially in the upper portion of the Sani Pass profile, and these cores usually show a marked deviation from the average D and I values. There is therefore some doubt whether in these cases the magnetization is the original TRM. The possibility exists that these abnormally high J values may be due to a high TRM owing to chemical or physical differences in the magnetic constituents of these rocks. It was thought that further light would be thrown on this point by consideration of the ratio of the NRM to the artificial TRM acquired by cooling the rock through its Curie point in the Earth's field. Accordingly, four specimens from each core were heated in a nitrogen atmosphere to 650 °C and then allowed to cool in the Earth's field. The details of the procedure are described in Part I of this study (van Zijl, Graham & Hales 1962). The ratios NRM/TRM are shown in Figure 2.

The fact that a few irregularities in the J curves of Figure 1 (e.g. a peak at 710 ft in the Maseru area) have been eliminated from Figure 2 indicates that there

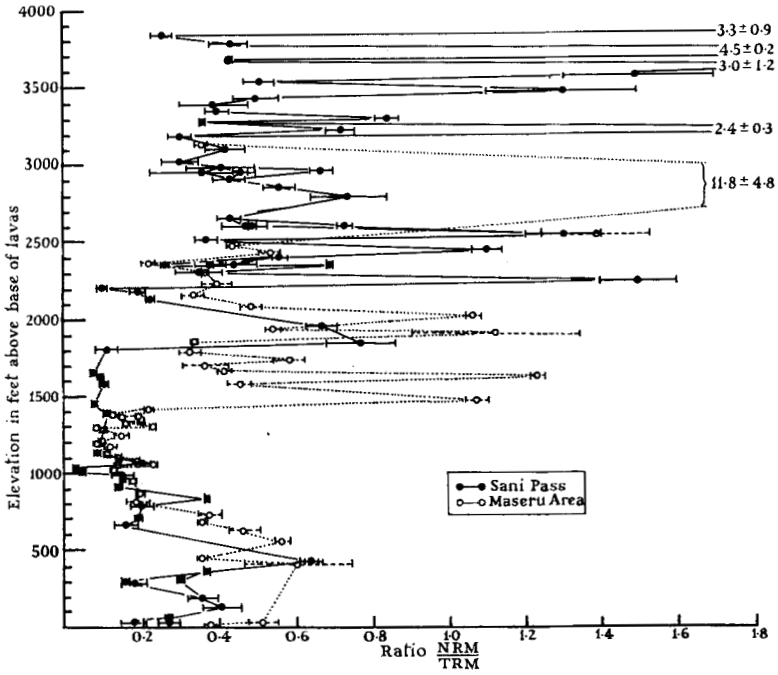


FIG. 2.—The ratio NRM/TRM and standard deviation, plotted against the elevation above the base of the lavas for cores from Sani Pass and the Maseru area. In some cases a particular lava flow was sampled by more than one core.

are local differences in the magnetic composition of these lavas. However, the remarkable similarity of Figures 1 and 2 strongly suggests an overall uniformity in the magnetic character of these rocks. This is confirmed in the ore-microscope studies and the X-ray analyses reported in Part I.

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It will be seen that the abnormally high peaks remaining in Figure 2 correspond with high peaks in the J curve (Figure 1(a)). Bearing in mind the mean standard deviations of these irregular peaks with limits well out of the average range of their neighbours it is reasonable to assume that the NRM of these samples is not entirely due to TRM.

(*c*) *A.C. demagnetization.*—In addition to the TRM component, the NRM of igneous rocks can consist of an isothermal remanent magnetization (IRM) acquired by exposure for long periods of time in the Earth's magnetic field, or during lightning discharges, and a remanent magnetization due to post-consolidation chemical changes in the magnetic constituents of the rock. Many workers have successfully used a.c. fields to remove such secondary magnetization from rocks, e.g. Creer (1958). Further, the form of the a.c. demagnetization curves can give very useful information about the nature of the remanent magnetism of the sample (Uyeda 1958, Rimbert 1958, Cox 1961, and many others).

Accordingly one specimen from each core was demagnetized at progressively increasing a.c. fields. The technique is similar to that used by Creer (1959) and the equipment used has been described by Graham (1961).

The results fall into three categories. A.C. demagnetization curves for typical specimens of each of the three categories are shown in Figure 3 labelled NRM. The a.c. demagnetization curves marked TRM are for specimens from the same cores. These were given an artificial TRM by cooling in the Earth's field. For Type 2 the figure shows also the a.c. demagnetization curve for a specimen given artificial IRM by d.c. magnetization. Stereographic plots of the changes of direction during demagnetization are also given in Figure 3.

Type 1. The decrease in intensity, J, is gradual, reaching a value of about 0·4 of the initial intensity at 250 Oe peak a.c. field. Furthermore, the directions of magnetization remain unchanged. The drop in intensity compares well with that of a similar specimen from the same core with pure artificial TRM. This type may therefore be classed as pure natural TRM.

Type 2. In this case the initial NRM is characteristically very high and decreases very rapidly at the early stages of a.c. demagnetization. The curve gradually flattens out and at 200 Oe demagnetizing field reaches an intensity of about 5×10^{-4} gauss which may be as little as 0·03 of the original intensity. The abnormally high initial NRM is many times greater than the NRM for neighbouring cores. The original directions of magnetization for specimens in this category from the same core generally show scattered directions. In addition, the direction of magnetization changes rather rapidly with increasing a.c. field, but as can be seen from Figure 3 an "endpoint" is never reached. It is clear that this type of magnetization is very different from pure TRM but closely resembles the IRM artificially produced by a d.c. field in a specimen from the same core. It may be noted that these cores invariably correspond to the high J values and the high NRM/TRM ratio in Figures 1 and 2 respectively.

It has been shown by Cox (1961) and Graham (1961) that specimens with abnormally high NRM and with a.c. demagnetization curves very similar to those presented here, have been remagnetized by lightning and it seems probable therefore that specimens of Type 2 have been remagnetized in this manner.

Because a.c. "washing" often failed to restore completely the original direction of magnetization of specimens with abnormally high magnetization, thermal demagnetization was attempted. This procedure also failed to remove the secondary magnetization.

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Type 3. Although the initial NRM is not very different from that of Type 1, it changes rapidly in the early stages of a.c. demagnetization. The intensity of normally magnetized specimens decreases whereas that of reversely magnetized samples increases initially. From about 100 Oe onwards the curves for both normally and reversely magnetized samples closely follow that of Type 1. The change in the direction of magnetization with increasing a.c. field is small and ceases at about 100–150 Oe. A small viscous magnetization, which is always in the direction

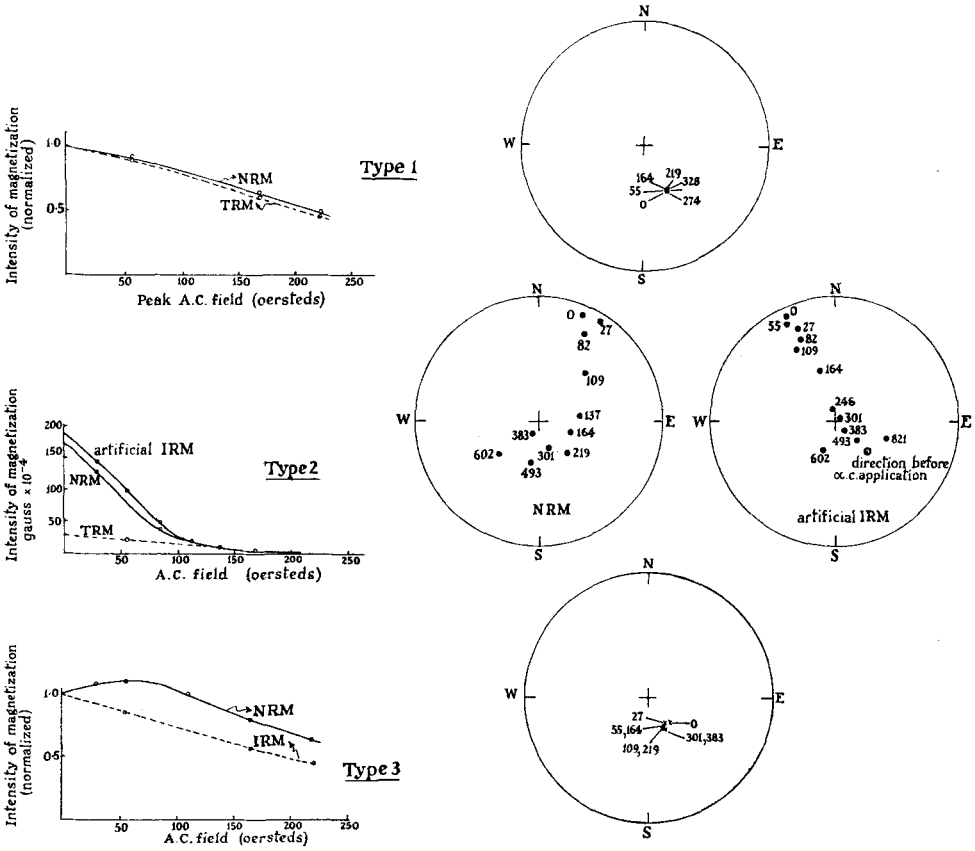


FIG. 3.—A.c. demagnetization curves and stereographic plots of the changes in directions of magnetization of three types of remanent magnetizations present in the lavas.

Stereograms: Lower hemisphere plots.

● South poles down. × North poles down.

164—value of peak a.c. field in oersteds.

of the present field, superimposed on stable TRM would account for these observations (Thellier 1938).

It should be noted that a number of a.c. demagnetization curves indicated the presence of a secondary magnetization which is of the same order of magnitude as that of Type 3. Although in these cases it was not possible to show directly that the direction of such magnetization was in the direction of the Earth's field, the important point is that these magnetizations could be eliminated at small a.c. fields. The origin of the latter spurious magnetizations is tentatively also assumed to be caused by a lightning strike at some distance away from the sample.

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3. Final results

The above considerations have shown that in addition to a stable magnetization of TRM type, there exists in these rocks two other types of magnetization. The one characterized by initial abnormally high NRM and thought to be due to lightning cannot be removed effectively by a.c. demagnetization because although the directions of magnetization change with increasing a.c. field, they start scattering before reaching an "endpoint". Specimens artificially magnetized (by a d.c. field) to similar intensities, show exactly the same phenomenon. Fortunately, however, this type of anomalous magnetization can generally be recognized from the form of the a.c. demagnetization curves (Uyeda 1958) and the changes in direction of magnetization at fairly low a.c. fields.

The milder type of magnetization, thought to be either viscous magnetization or weak IRM, is easily eliminated at low a.c. fields.

For these reasons it was decided to carry out the following series of measurements:

(i) One typical specimen from each core was demagnetized in successive steps to 219 Oe.

(ii) The curves from (i) were compared with those obtained by demagnetizing a similar specimen which had been given an artificial TRM in the Earth's field.

(iii) Three other typical specimens from each core were demagnetized at 219 Oe.

It was expected that this series of measurements would make it possible to identify the type of magnetization by studying the forms of the demagnetization curves (i) and (ii). In particular, if the four specimens from a core showed scatter after demagnetization, then the results from that core were disregarded both for the determination of the mean D and I, and in the discussion of the intensities. The low intensity, relatively "soft" magnetization should be eliminated at 219 Oe and moreover four specimens from the same core demagnetized at this level should show tight grouping.

The a.c. demagnetization curves for the specimens having only natural TRM should resemble those of similar specimens from the same core given an artificial TRM. Four specimens (having NRM of TRM type) from the same core having only pure natural TRM should remain tightly grouped without changing direction. Any "soft" component of magnetization which may be too small to be detected in the a.c. demagnetization curves will also be removed at this level

The results after this treatment are shown in Figure 4. In this figure $(NRM/TRM)_{219\ Oe}$, i.e. the ratio of the NRM of the specimens after demagnetization at 219 Oe to the magnetization of a specimen from the same core given an artificial TRM by cooling in the present field and then demagnetized at 219 Oe, is plotted against height in feet above the base of the lavas. Those cores which showed scatter have been excluded from the curves. Measurements on these cores are indicated by crosses.

In the curve showing the $(NRM/TRM)_{219\ Oe}$, the effects of any viscous or mild IRM component of magnetization have been removed. Also, those samples in which stronger IRM which could be recognized but which could not be completely removed have been excluded. However, it is still possible that some of the very large peaks could be due to very high intensity IRM (probably due to lightning) in spite of the fact that it is not appreciably altered by relatively small a.c. fields.

In discussing the intensity of the ancient field below, the ratio of $(NRM/TRM)_{219\ Oe}$ is used. The decrease in intensity of the natural TRM in the a.c. field

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of 219 Oe is compensated by the similar decrease of the artificial TRM. However, it was possible that the acquisition of secondary magnetization and its subsequent removal might have affected the original TRM component.

Figure 5 shows the average a.c. demagnetization curves of four specimens having artificial TRM, from each of two normal cores, two reversed cores and two transition cores. In addition, another specimen from the same core was given an artificial TRM and then magnetized in a d.c. field to produce a magnetization about

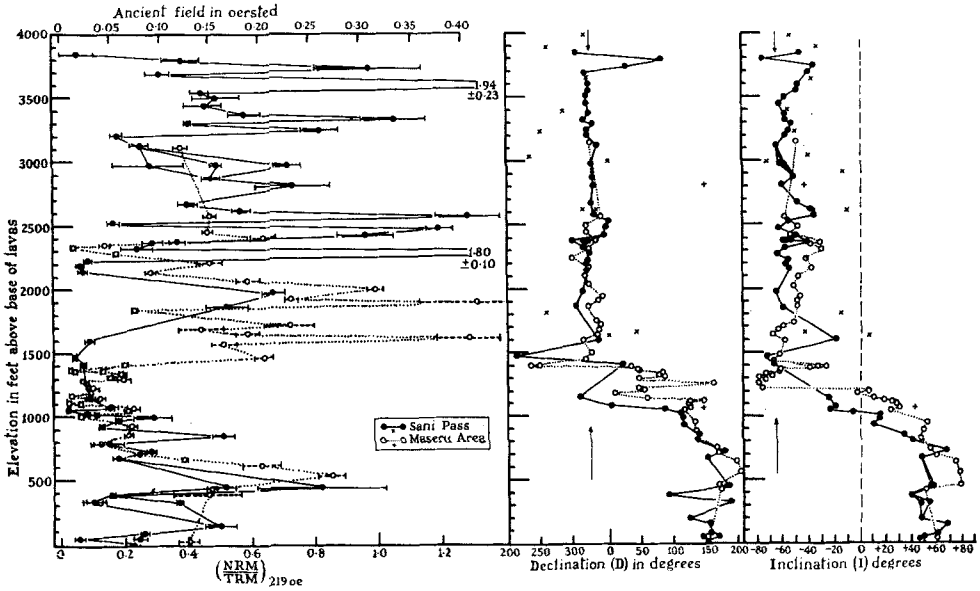


FIG. 4.—The ratio $(NRM/TRM)_{219\,Oe}$ and standard deviation, and declination and inclination after A.C. treatment, plotted against height in feet above the base of the lavas. The arrows indicate the declination and inclination of the present field.

two to four times that of the artificial TRM. The specimens were then demagnetized.

It will be observed that the demagnetization curves of the specimens of the latter group approach asymptotically those of the corresponding member of the first group.

The authors regard these observations as justifying the use of the ratio $(NRM/TRM)_{219\,Oe}$, in the discussion of intensities.

4. Discussion

There is a remarkable correspondence in the *D* and *I* curves for the two areas especially in the transition zone. The time covered by the latter is represented by the extrusion of about 600 ft of lava (from 800–1 400 ft above the base of the succession). It may be noted that the transition zone in the Maseru area is about 200 ft higher above the lava base than at Sani Pass. In the period of transition it is clear that the ratio $(NRM/TRM)_{219\,Oe}$ is appreciably less than its average value in the normal and reversed zones.

Königsberger (1938) attempted to determine the magnitude of the ancient magnetic field by considering the ratio of the NRM and artificial TRM acquired

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by heating the rock to above Curie point(s) of the magnetic constituent(s) and cooling it in a known field. The ore microscopic study and X-ray analysis described in Part I, did not reveal any change in the ore-minerals during heating. It is unlikely therefore that the artificial TRM being used as a reference arises as a result of chemical change in the magnetic materials. The use of the ratio $(NRM/TRM)_{219\ Oe}$ instead of (NRM/TRM) means that the effect of all soft components has been removed.

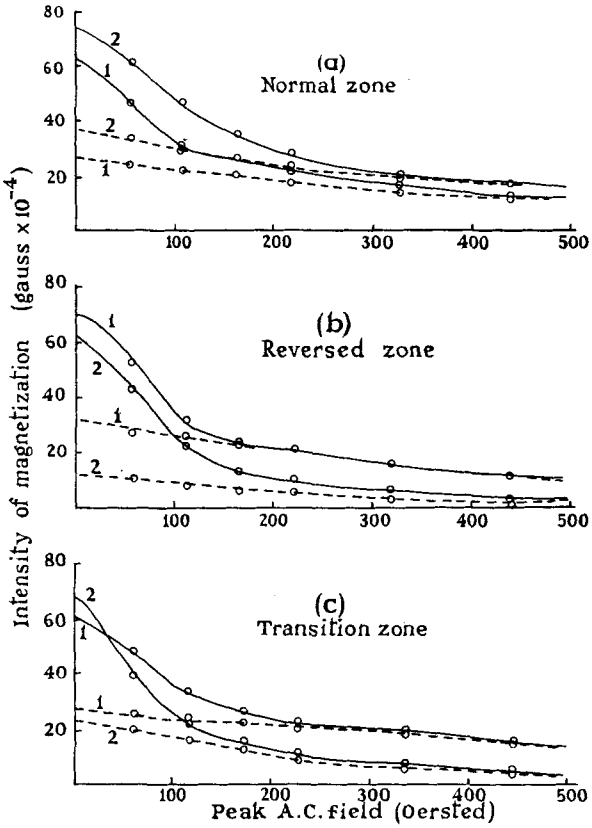


FIG. 5.—A.c. demagnetization curves for specimens with artificial TRM and specimens from the same respective core with artificial IRM from (a) the normal zone, (b) the reversed zone, and (c) the transition zone.
- - - - - specimens with only artificial TRM.
——— specimens with both artificial TRM and IRM.

In the $(NRM/TRM)_{219\ Oe}$ curves in Figure 4 the value of the ancient field has been indicated on the top scale. The values have been determined from the fact that at low fields ($< 0{\cdot}8$ Oe) the TRM acquired by a rock when cooled through the Curie point(s) of the magnetic constituent(s) varies linearly with the magnitude of the small applied field (Nagata 1953, Wilson 1961, etc). The ancient magnetic field obtained in this manner has not been corrected for any decay of the TRM throughout geological time.

(a) *The normal zone.*—The effects of the secular variation in this zone are rather pronounced. The tendency is for the declination to remain reasonably constant whereas the inclination shows periodically rather large departures from the typical average value, e.g. at 2 300 ft and 2 500 ft in the Maseru area and at Sani Pass

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respectively. At these elevations the intensity of the field decreases sympathetically. This effect is particularly pronounced in the Maseru area.

The average for D and I (calculated vectorially) over the entire normal zone, i.e. from 1 800–3 800 ft at Sani Pass and from 1 400–3 100 ft in the Maseru area, are given in Table 1 together with the calculated positions of the south geomagnetic pole in present day co-ordinates assuming a dipolar field. The table also shows the 99 per cent confidence limits (Fisher 1953). In the calculation of D, I and α each core was given unit weight.

Table 1

Average D and I, α 99 per cent, position of south geomagnetic pole, δp 99 per cent, δm 99 per cent and standard deviation of directions for the normal zone at Sani Pass and in the Maseru area

	Sani Pass	Maseru
Average D	329° 22′	329° 29′
Average I	−53° 31′	−50° 25′
α 99 per cent	4° 5′	6° 43′
Geomagnetic S pole:		
longitude	101° 33′ E	106° 23′ E
latitude	63° 41′ S	63° 41′ S
δp 99 per cent	3° 58′	6° 3′
δm 99 per cent	5° 41′	9° 1′
$\delta = (\cos^{-1} R/N)$*	10° 19′	13° 3′
N (number of samples)	33	21

* $\delta = (\cos^{-1} R/N) = 15°$ for present field (Vestine & others 1947).

In the table the standard deviations δ of single observations are given. These standard deviations are of the same order as those found by Vestine (1947) for the present day variations of the Earth's field from the axial dipole field.

As shown in Figures 6(a) and 6(b) the circles of confidence for the directions of magnetization and the ovals of confidence of the poles, as determined from the two areas, overlap. Furthermore the mean direction of magnetization of 12 samples of Karroo basalts from Rhodesia reported by Nairn (1960) gives a pole position not significantly different from those found from the normal zone. In view of the fact that the Basutoland lavas are separated from those in Rhodesia by some 800 miles, those results support the hypothesis of an average axial dipole field.

Graham & Hales (1957) in their investigation of the Karroo dolerites report average values of D and I for the normal dolerites of Winkelhaak and Estcourt. The values for the Winkelhaak and Estcourt areas differ but the difference between these values and those of the present study are not significant having regard to the fact that the number of independent observations at Winkelhaak and Estcourt is small.

The average ratios of NRM/TRM at 219 peak a.c. Oe demagnetization for normal zone cores from Sani Pass and the Maseru area agree very well and are $0{\cdot}596 \pm 0{\cdot}081$ and $0{\cdot}609 \pm 0{\cdot}106$ respectively which correspond to a field strength of 0·19 Oe (see NRM/TRM curve—Figure 4), assuming that there has not been any decay of TRM with time.

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Assuming a dipole field of the same moment as the present field, the field strength at the sample sites can be determined from the average I values by means of the formulae

$$H = \frac{P}{R^3}(4 - 3\cos^2\theta) \quad \text{and} \quad \tan I = 2\tan\theta,$$

where P = mean present day dipole moment
R = present day radius of the Earth
θ = geographic latitude
I = magnetic inclination.

The value of H thus found is 0·42 Oe. The large difference between this value and that of 0·19 Oe obtained from the NRM/TRM ratio could be due to

(i) The radius of the Earth in Triassic–Jurassic time being 30 per cent larger than it is today. There is no evidence from surface geology that a contraction of this order has occurred during the last 150–200 million years.

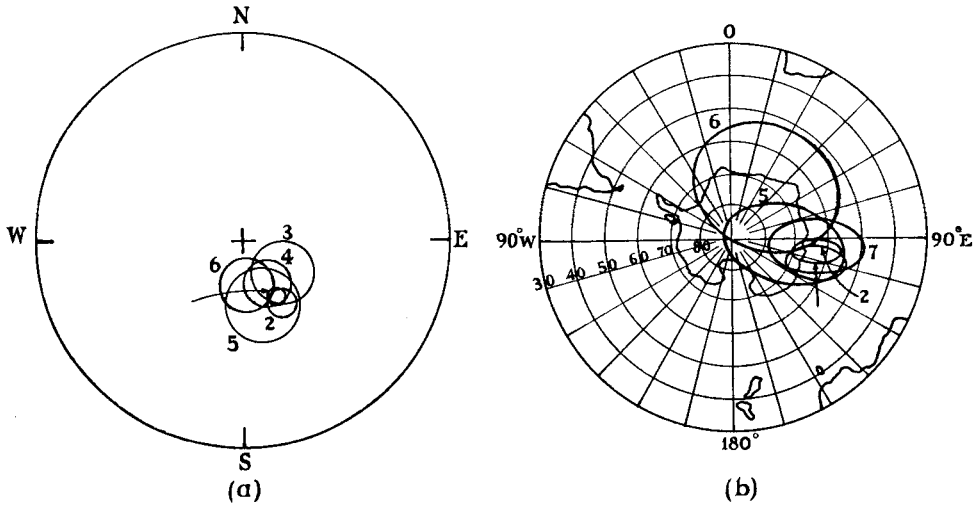


FIG. 6.—(a) The circles of 99 per cent confidence for the mean directions of magnetization of the Karroo lavas and dolerites, (b) The ovals of 99 per cent confidence for the pole positions.

1. Sani Pass normal zone.
2. Maseru Area normal zone.
3. Winkelhaak normal dolerites
4. Estcourt normal dolerites
   } Graham & Hales (1957).
5. Sani Pass reversed zone.
6. Maseru Area reversed zone.
7. Karroo lavas, Rhodesia (99 per cent confidence ovals, calculated from Cox & Doell, 1960; measurements by Nairn, 1957).

(ii) the dipole moment of the ancient field being smaller than it is today.

(iii) the loss of TRM with time in these rocks having been large (≒ 50 per cent).

There is no satisfactory method of deciding between (ii) and (iii). The results nevertheless seem to indicate that if the large decrease in intensity of the Earth's field over the last 2 000 years is valid (Thellier & Thellier 1959), this trend did not persist from the Jurassic.

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(b) *The reversed zone.*—It is unfortunate that the reversed zone occupies only about 700ft of the lava succession as against 2000ft for the normal zone at Sani Pass. It would not be expected that the effects of secular variation can be meaned out over such a short vertical distance particularly as so few cores are available for analysis. Additional information cannot be obtained from the underlying Cave Sandstone as this aeolian sandstone is magnetically unstable and chemical changes may have occurred during baking by the overlying lava (see Part I).

Table 2 shows the average D and I values and the pole positions with the same statistics as in Table 1. In Figure 6(*a*) and 6(*b*) it is seen that at the 99 per cent level the differences in the average directions of magnetization and the pole positions respectively are not significantly different for the reversed zones in the two areas. It should, however, be noted that at those confidence limits there is a significant difference between the normal and reversed zone in the Maseru area, disregarding sign. As for the normal zone, the standard deviations of single observations for the departure of the ancient field from the axial dipole field is comparable with the value of 15° for the present Earth's field.

Table 2

Average D and I, α 99 per cent, position of north geomagnetic pole, δp 99 per cent, δm 99 per cent and standard deviation of directions for reversed zone at Sani Pass and in the Maseru area

	Sani Pass	Maseru
Average D	162° 12′	177° 30′
Average I	+54° 6′	+65° 54′
α 99 per cent	18° 54′	13° 42′
Geomagnetic N pole:		
longitude	96° 6′E	33° 0′E
latitude	74° 6′S	71° 6′S
δp 99 per cent	13° 18′	18° 16′
δm 99 per cent	18° 56′	22° 22′
$\delta = (\cos^{-1} R/N)$	18° 56′	12° 36′
N(number of samples)	13	7

The $(NRM/TRM)_{219\,Oe}$ curves in Figure 4 show that the average level of the ratio in the reversed zone is greater than that in the transition zone, but the contrast is not as great as the difference between the levels of the ratio NRM/TRM in the normal and transition zones. It is possible that the reversed field was already changing towards the transition when the first out-pourings of lava occurred.

(c) *The transition zone.*—The $(NRM/TRM)_{219\,Oe}$ curves in Figure 4 for the two areas are remarkably similar especially in the transition zone. Bearing in mind the standard deviations of the mean ratios shown on the curves, this constitutes strong evidence for believing that the intensity of the ancient field decreased during the transition.

The ratio $(NRM/TRM)_{219\,Oe}$ for the transition zone at Maseru where detailed sampling was possible is $0{\cdot}123 \pm 0{\cdot}015$. The mean calculated was from the measurements on 22 cores covering 570ft of the lava succession. For the normal zone the mean value of the ratio $(NRM/TRM)_{219\,Oe}$ is $0{\cdot}609 \pm 0{\cdot}106$. The 99 per cent confidence limits for the means for the transition and the normal zones are $\pm 0{\cdot}039$ and $\pm 0{\cdot}274$ respectively. The values therefore differ significantly at the 99 per

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cent confidence level. The normal zone values include three cores, *viz.* at 2 530, 1 880 and 1 610 ft above the base of the lavas, in which the ratio $(NRM/TRM)_{219\ Oe}$ is unusually high. If these three high values are excluded the mean $(NRM/TRM)_{219\ Oe}$ for the normal zone is $0{\cdot}451 \pm 0{\cdot}072$; the 99 per cent confidence limits being $\pm 0{\cdot}186$. Thus, although the mean has decreased the 99 per cent confidence limits have decreased correspondingly and the values of $(NRM/TRM)_{219\ Oe}$ for the normal and transition zones are still significantly different. On this basis it is suggested that the field intensity decreased by a factor of about four to five during the transition.

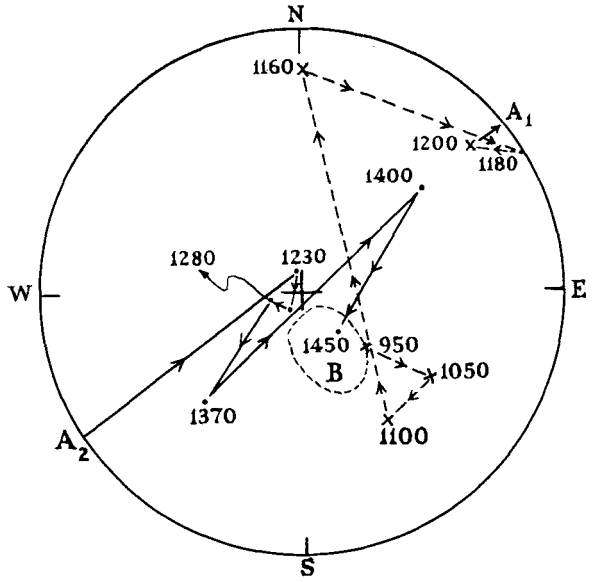


FIG. 7.—A lower hemisphere stereographic plot of the directions of magnetization of flows from the transition zone in the Maseru area.
● South poles downwards. × North poles downwards.
A_2 follows on A_1.
B represents the region occupied by typical specimens from the normal and reversed zones. The numbers indicate the elevation in feet of the flow above the base of the lavas.

Sigurgeirson (1957), working on a reversal in the Icelandic basalts, indicated that there was possibly a decrease in intensity by a factor of two during the transition. There were, however, insufficient data for a satisfactory analysis.

A remarkable feature of the curves is that just before and after transition the field intensity seems to build up to an unusually high value.

During the transition the *D* and *I* values change rapidly in a series of jerks from a reversed to a normal direction as shown in Figure 7 for the Maseru area, where sampling of every flow in the transition zone was possible. The tendency is for the direction to change rapidly to a point where there is a slight pause to execute a clockwise circular rotation before a rapid change to a new centre of rotation is made.

Brynjólfssen (1957) has observed the same clockwise type of variation in the reversal of the Icelandic basalts. Chevallier (1925) was able to trace the secular variation for historical lava flows on Mount Etna from 1200 A.D. onwards and the plots show the same clockwise rotation. Runcorn has recently (1955) noted that this

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clockwise rotation of the secular variation is explainable in terms of the westward drift of the geomagnetic field. It is estimated that a complete clockwise revolution of the secular variation takes roughly 1 000–3 000 years. This being so, the maximum time taken for the field reversal in the Stormberg would be 12 000 years, a figure which does not differ appreciably from Hosper's and Roche's estimate of 10 000 years (Jacobs, Russell & Wilson 1959, pp. 135). Unfortunately it is not possible to make an order of magnitude estimate of the time covered by the transition zone from geological field evidence. In the Maseru area there are 24 flows in the transition zone with an average thickness of about 15 ft. One such flow consolidates in less than 10 years (Jaeger 1957). The minimum estimate, therefore, for the time of transition is less than 150–200 years. Weathering between flows is absent but this does not necessarily mean that the flows followed one another rapidly. There is evidence from an interbedded aeolian sandstone 1 100 ft up in the lava succession that the environment was arid. In such a climate weathering is negligible.

Elsasser (1956) has proposed that the predominant dipole component of the Earth's field decreases to zero during a field reversal. This study supports this view and the rapid movement of the field direction in the transition period seems to indicate that the non-dipole components are predominant.

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